Characteristic Poly. & Minimal Poly.

linear operator : $T: V \to V$

(? def ?)

Eigenvalues & Eigenvectors

Def:

$T(\underline{v}) = \lambda \underline{v} \iff \underline{v} \in \ker(T - \lambda I)$

$\hookrightarrow$ $\underline{v}$: an eigenvector of T with eigenvalue λ

Characteristic poly.

Def.: the char. poly. $\chi_T(x)$ of $T: V \to V$

is $\chi_T(x) := \det\left((T)_{\mathcal{A}\mathcal{A}} - xI\right)$

($\mathcal{A} = \{\underline{v_1}, \cdots, \underline{v_n}\}$ is a basis of V)

?

Remark : If we use $\mathcal{B} = \{\underline{v_1'}, \cdots, \underline{v_n'}\} \neq \mathcal{A}$

then $T_{\mathcal{A}\mathcal{A}} \neq T_{\mathcal{B}\mathcal{B}}$

but $\det\left((T)_{\mathcal{A}\mathcal{A}} - xI\right) = \det\left((T)_{\mathcal{B}\mathcal{B}} - xI\right)$

so $\chi_T(x)$ does not depend on choice of basis

(? A逆知, B知)

证: $T_{\mathcal{A}\mathcal{A}} = \mathcal{C}_{\mathcal{B}\mathcal{A}}^{-1} T_{\mathcal{B}\mathcal{B}} \mathcal{C}_{\mathcal{B}\mathcal{A}}$ Let $C = \mathcal{C}_{\mathcal{B}\mathcal{A}}$

$\det(T_{\mathcal{A}\mathcal{A}} - xI) = \det(C^{-1} T_{\mathcal{B}\mathcal{B}} C - x C^{-1} I C)$

$= \det(C^{-1}(T_{\mathcal{B}\mathcal{B}} - xI)C) = \det(C^{-1}) \det(T_{\mathcal{B}\mathcal{B}} - xI) \det(C)$

$= \frac{1}{\det(C)} \det(\sim) \det(C) = \det(\sim)$

E.g.: $T:\mathbb{R}^2\to\mathbb{R}^2$, $T\begin{pmatrix}\alpha\\ \beta\end{pmatrix} := \begin{pmatrix}1&2\\3&4\end{pmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix}$,

$\mathcal{B}=\left\{\begin{pmatrix}1\\0\end{pmatrix},\begin{pmatrix}0\\1\end{pmatrix}\right\} \Rightarrow T_{\mathcal{B}\mathcal{B}} = \begin{pmatrix}1&2\\3&4\end{pmatrix}$

then $\chi_T(x) = \det(T_{\mathcal{B}\mathcal{B}} - xI)$

$= \det\begin{pmatrix}1-x&2\\3&4-x\end{pmatrix} = x^2-5x-2$

(★!) → $I:\mathbb{R}^2\to\mathbb{R}^2$, $I=\begin{pmatrix}1&0\\0&1\end{pmatrix}$

(Ex. 证 $A=\left\{\begin{pmatrix}1\\1\end{pmatrix},\begin{pmatrix}1\\-1\end{pmatrix}\right\}$

(接) 能通过 解 $\chi_T(x)=0$

来找到所有 eigenvalues / eigenvectors of $T:V\to V$

过程:

Suppose $\chi_T(\lambda)=0$

$\iff \det(T_{AA}-\lambda I)=0$

(In MAT2040) $\iff \exists\, \underline{w}\in\mathbb{F}^n\setminus\{\underline{0}\}$,

s.t. $T_{AA}\underline{w} = \lambda \underline{w}$

(Let $\underline{v}\in V$, s.t. $[\underline{v}]_A = \underline{w}\in\mathbb{F}^n$)

$\iff [T(\underline{v})]_A = \lambda[\underline{v}]_A$

$\iff [T(\underline{v})-\lambda\underline{v}]_A = 0_{\mathbb{F}^n}$

$\iff \underline{T(\underline{v})=\lambda\underline{v}}$

i.e., $\underline{v}$ is an evector of T with evalue λ

② L. Op. Induced from Poly. (?)

Def.: $T: V \to V$ is a L. op.,

$p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \in \mathbb{F}[x]$

define $p(T) = a_n T^n + \dots + a_2 T^2 + a_1 T + a_0 I$

$T^n = \underbrace{T \circ T \circ \dots \circ T}_{n \text{ times}}$, $T^2 = T \circ T$ $: V \to V$

$I : V \to V$ identity operator

E.g.: $T: \mathbb{R}^2 \to \mathbb{R}^2$, $T\begin{pmatrix}\alpha \\ \beta\end{pmatrix} = \begin{pmatrix}1 & 2 \\ 3 & 4\end{pmatrix}\begin{pmatrix}\alpha \\ \beta\end{pmatrix}$

$f(x) = x^2 + 4$,

then $f(T) = \begin{pmatrix}1 & 2 \\ 3 & 4\end{pmatrix}^2 + 4I = \begin{pmatrix}1 & 2 \\ 3 & 4\end{pmatrix}\begin{pmatrix}1 & 2 \\ 3 & 4\end{pmatrix} + 4\begin{pmatrix}1 & 0 \\ 0 & 1\end{pmatrix} = \begin{pmatrix}11 & 10 \\ 15 & 26\end{pmatrix}$

$\chi_T(T) =$ (?) (?)

$= \begin{pmatrix}0 & 0 \\ 0 & 0\end{pmatrix}$

Remark:

Compute $(f(T))_{\mathcal{B}\mathcal{B}}$ for $\mathcal{B} = \left\{\begin{pmatrix}1 \\ 0\end{pmatrix}, \begin{pmatrix}0 \\ 1\end{pmatrix}\right\}$ in $\mathbb{R}^2$

$f(T) = T^2 + 4I = T \circ T + 4I : V \to V$

Then $(f(T))_{\mathcal{B}\mathcal{B}} = (T \circ T)_{\mathcal{B}\mathcal{B}} + 4(I)_{\mathcal{B}\mathcal{B}}$

$= (T)_{\mathcal{B}\mathcal{B}} \cdot (T)_{\mathcal{B}\mathcal{B}} + 4\begin{pmatrix}1 & 0 \\ 0 & 1\end{pmatrix}$

$= \begin{pmatrix}1 & 2 \\ 3 & 4\end{pmatrix}\begin{pmatrix}1 & 2 \\ 3 & 4\end{pmatrix} + 4\begin{pmatrix}1 & 0 \\ 0 & 1\end{pmatrix} = \begin{pmatrix}1 & 2 \\ 3 & 4\end{pmatrix}^2 + 4\begin{pmatrix}1 & 0 \\ 0 & 1\end{pmatrix}$

$(f(T))_{\mathcal{B}\mathcal{B}} = \begin{pmatrix}11 & 10 \\ 15 & 26\end{pmatrix}$ and $(\chi_T(T))_{\mathcal{B}\mathcal{B}} = \begin{pmatrix}0 & 0 \\ 0 & 0\end{pmatrix}$

$\Rightarrow \chi_T(T) : V \to V$, $\underline{v} \mapsto \underline{0}$ $\forall$ (maps all $\underline{v}$)

Cayley - Hamilton Theorem

$\chi_T(T) = \underline{0}_{V \to V}$ for all $T: V \to V$ $(\dim(V) < \infty)$

Minimal Poly.

Def. : $T: V \to V$

$m_T(x) \in \mathbb{F}[x]$ is a minimal poly.

$$\Leftarrow \begin{cases} m_T(x) \text{ is monic } (①) \\ m_T(T) = \underline{0}_{V \to V} \ (②) \\ m_T(x) \text{ is of smallest possible degree} \\ \qquad \text{satisfying } ① \& ② \ (③) \end{cases}$$

E.g. : $T: \mathbb{R}^2 \to \mathbb{R}^2$, $T\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$

$q(x) = x^2 - 5x^2 - 2$ (monic)

satisfies $q(T) = \underline{0}_{\mathbb{R}^2 \to \mathbb{R}^2}$,

$\deg(q(x)) = 2$

(Claim : $m_T(x) = q(x)$)

Suppose (on contrary),

$\exists$ monic poly. $s(x) = x - k$,

s.t. $s(T) = 0_{\mathbb{R}^2 \to \mathbb{R}^2}$ and $\deg(s) < \deg(q)$

However : (见题) (W.24)

1. For $T: V \to V$, $\dim(V) < \infty$,

$m_T(x)$ always exists

Proof:

Consider $X := \mathrm{Hom}(V, V)$

Then $\dim(X) = \dim(V) \cdot \dim(V) = n^2$

Consider $\underbrace{\{I, T, T^2, T^3, \cdots, T^{n^2}\}}_{(n^2+1)\text{个 operators}} \leq W$ $(\dim(W) = n^2)$

? is l. dep. ☆, i.e.: $\exists \alpha_0, \cdots, \alpha_{n^2}$ (? scalars) not all zeros ?,

$$\alpha_0 I + \alpha_1 T + \cdots + \alpha_{n^2} T^{n^2} = 0$$

☆ Suppose that the last nonzero term is α_m ?

Then $\alpha_0 I + \alpha_1 T + \cdots + \alpha_m T^m = 0_W$

$$\Rightarrow T^m + \frac{\alpha_0}{\alpha_m} I + \frac{\alpha_1}{\alpha_m} T + \cdots + \frac{\alpha_{m-1}}{\alpha_m} T^{m-1} = 0_W$$

? ☆ Consider ([illegible])